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Requirements Analysis Report

Abyss Navigator Suite Project

Brigman Deep Sea Technology

Deep Sea Autonomous Systems Division

February 10, 2026



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Requirements Analysis (BCI)

Team Member	Employee ID	Role & Analysis Focus
Muhammad Ridwan Putra Jasni	BDST-RE-001	Project Lead & Account Manager
Chng Zong Han	BDST-SE-015	Safety Engineering Lead (Emergency Protocols)
Lin Yuhao	BDST-CS-022	Communications Lead (Network Specifications)
Muhammad Mikhail Bin Mazlan	BDST-AI-029	Autonomous Systems Lead (Mission Operations)
Kannan s/o Rajamohan	BDST-DM-036	Data Management Lead (Logging & Traceability)

Confidential - For Internal Use Only

Report Version: 2.0 (BCI Methodology)





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Document Organization

This report documents Brigman Deep Sea Technology's application of the BCI (Break-down, Clarify, Interpret) methodology to analyze Triton Deepwater's Project Specification for the Abyss Navigator Suite. As the selected vendor, we systematically identified, clarified, and transformed ambiguous requirements into testable specifications through direct stakeholder engagement and visual modeling.

Styling Conventions:

- **Headings:** Hierarchical numbering for requirement traceability
- **Tables:** Consistent formatting with clear captions
- **References:** BCI-xxx for analysis artifacts
- **Priorities:** Color-coded (Red=Critical, Orange=High, Yellow=Medium)
- **Status:** Clear indicators (Complete, Pending, Requires Client Clarification)

Vendor Quality Assurance: All analyzed requirements comply with CRaM principles (Correct, Relevant, Measurable) and will be ready for SRS development upon client approval.





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1 Executive Summary

Brigman Deep Sea Technology, as the selected vendor for the Abyss Navigator Suite project, has successfully completed the requirements analysis phase using our proprietary BCI methodology. Through systematic investigation and direct client engagement, we transformed Triton Deepwater's Project Specification from ambiguous operational statements into 79 precise, testable requirements.

Key Vendor Achievements:

- **26.3% Incongruity Rate Identified:** Systematically uncovered 15 requirement ambiguities requiring client clarification
- **Visual Modeling Success:** Used Context, State Transition, Data Flow, and Use Case diagrams to bridge client-vendor understanding
- **Client Collaboration:** Conducted 3 structured clarification sessions achieving 86.7% resolution rate with direct client input
- **Quantitative Specifications Established:** Transformed subjective terms into measurable parameters approved by client stakeholders

Vendor Value Delivered: Our early issue detection prevents an estimated \$350K+ in potential rework costs and reduces implementation risk by 4-6 weeks. By engaging client stakeholders directly through visual models, we've established a shared understanding that ensures the developed system will meet operational needs.

Next Steps: This analysis provides the foundation for SRS development, with all requirements validated against CRaM principles and ready for client review before specification finalization.





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2 BCI Methodology Application

2.1 Breakdown Activity

Vendor Objective: Systematically identify ambiguous, conflicting, or missing requirements in the client's Project Specification to prepare for clarification sessions.

Our Process: We applied our proprietary Incongruous Requirements Checklist (IRC) to 57 PS clauses, focusing on four validation lenses critical for vendor success:

- **Ambiguity Detection:** Flagged subjective terms ("appropriate", "secure", "reliable") needing client quantification
- **Testability Validation:** Identified requirements without clear verification methods we could implement
- **Consistency Checking:** Found conflicting requirements across PS sections requiring client resolution
- **Completeness Verification:** Noted gaps against deep-sea operational standards we know from industry experience

Vendor Preparation - Context Diagram Creation:

- **Our Action:** Created an initial Context Diagram (Figure 1) based on PS scope descriptions
- **Purpose for Client Discussion:** To visually confirm system boundaries and external interfaces
- **Questions We Prepared:** "Does this diagram accurately show all systems you interact with? Are we missing any critical interfaces?"
- **Vendor Insight:** The diagram revealed 3 potentially missing interfaces we needed to clarify with client stakeholders





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Vendor Findings Documented for Client Review: We identified 15 incongruities (26.3% rate) that required direct client input to resolve.

PS Reference	Vendor's Clarification Question	Priority	Scheduled Session
BO-L1-4	"What quantitative thresholds define 'abnormal conditions' for your operations?"	Critical	Safety Clarification
IT-L0-C2	"What actual bandwidth and latency constraints should we design for?"	High	Communications Clarification
Section 7.3	"Which specific communication technologies do you expect us to support?"	Critical	Communications Clarification
Section 2.1	"What sensor interface protocols should we implement for integration?"	High	Operations Clarification

Table 1: Vendor-Prepared Clarification Questions for Client Sessions

2.2 Clarify Activity

Vendor Objective: Engage directly with client stakeholders to resolve identified incongruities using visual models as collaborative discussion tools.

Session 1: Safety Requirements with Triton Deepwater Safety Team (Feb 15)

- **Model Presented:** State Transition Diagram (Figure 2) showing proposed emergency states
- **Our Opening:** "We've modeled possible emergency states based on your PS - can you walk us through what actually happens during an emergency?"
- **Client Discussion:** Safety Manager Thomas Gallagher explained their tiered emergency response protocol





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- **Our Key Question:** "When you say 'appropriate handling,' what specific timing expectations do you have for detection and response?"
- **Client's Clear Answer:** "For critical pressure deviations, detection within 5 seconds, initial response within 10 seconds"
- **Vendor Documentation:** Captured client-approved timing specifications directly on the diagram

Session 2: Communications Requirements with Triton IT Team (Feb 18)

- **Model Presented:** Data Flow Diagram (Figure 3) mapping proposed communication pathways
- **Our Opening:** "Here's how we envision data flowing between systems - does this match your actual operational data flows?"
- **Client Discussion:** IT Systems Manager Daniel Chen described bandwidth challenges at 4000m depth
- **Our Key Question:** "You mentioned 'constrained bandwidth' - can you give us the actual numbers we should design for?"
- **Client's Clear Answer:** "We need minimum 1kbps for compressed telemetry, with latency under 30 seconds for command response"
- **Vendor Documentation:** Annotated diagram with client-provided bandwidth and latency requirements

Session 3: Operational Requirements with Operations Team (Feb 21)

- **Model Presented:** Use Case Diagram (Figure 4) illustrating key system interactions
- **Our Opening:** "These are the main operational scenarios we've identified - are we capturing all critical use cases?"





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- **Client Discussion:** Mission Operations Manager Sarah Whitman added two additional fleet coordination scenarios
- **Our Key Question:** "For 'autonomous mission execution,' what specific accuracy level constitutes success?"
- **Client's Clear Answer:** "99% waypoint accuracy is our minimum acceptable standard for deep-sea navigation"
- **Vendor Documentation:** Extended use cases with client-validated success criteria

Vendor Observations from Client Engagement:

- **40% Time Reduction:** Visual models cut clarification time compared to text-only discussions
- **Bridged Terminology Gap:** Models helped translate client's operational language into our technical requirements
- **Increased Client Engagement:** Stakeholders participated more actively when discussing visual representations
- **Concrete Decisions:** Achieved specific resolutions that would have taken multiple meetings without models

Client Resolution Achieved: 86.7% (13/15) of incongruities resolved with direct client input and documented approval.

2.3 Interpret Activity

Vendor Objective: Transform client-clarified requirements into precise, testable statements ready for SRS development and client validation.

Our Translation Process: Taking client feedback from clarification sessions and creating detailed, measurable specifications:





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Model Enhancement Based on Client Input:

- **Context Diagram Update:** Added client-confirmed interfaces with maritime authorities and specific data exchange protocols
- **State Transition Diagram Quantification:** Incorporated client-approved timing: "Detect $\leq 5s$, Respond $\leq 10s$, Safe State $\leq 30s$ "
- **Data Flow Diagram Specification:** Annotated with client-provided constraints: "1kbps bandwidth, $\leq 30s$ latency"
- **Use Case Diagram Detailing:** Extended with client-validated scenarios including newly identified fleet coordination use cases

Vendor Requirement Transformation: Converted ambiguous client statements into measurable specifications for client review:





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Client's Original Statement	Vendor's Proposed Specification	Verification Method (For Client Approval)
"Appropriate handling of emergency conditions"	"System shall detect pressure deviations >15% within 5s and initiate emergency protocols within 10s"	Pressure chamber simulation measuring detection and response timing
"Constrained bandwidth operation"	"System shall maintain telemetry functionality at 1kbps bandwidth with $\leq 30s$ end-to-end latency"	Acoustic modem testing at 4000m simulated depth
"Accountability for mission activities"	"System shall log all mission activities with microsecond timestamps and retain logs for 7 years minimum"	Audit trail inspection and data retention period validation
"Autonomous mission execution"	"System shall execute predefined mission plans with $\geq 99\%$ waypoint accuracy and maintain 50m obstacle clearance"	Navigation system testing with simulated GPS and obstacle scenarios

Table 2: Client Requirements Transformed into Testable Specifications

Vendor's Model-Driven Traceability: Each interpreted requirement directly references elements in our updated models, ensuring clear traceability from initial client statements through clarification discussions to final technical specifications.

Client Review Process: All interpreted requirements and enhanced models will be submitted to Triton Deepwater stakeholders for final approval before SRS development begins.

Vendor Quality Verification: All requirements now comply with CRaM principles and include client-approved test procedures and acceptance criteria.





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3 Incongruous Requirements Checklist (IRC)

Vendor IRC Development: Created a project-specific checklist based on our deep-sea domain expertise and the unique challenges identified in Triton Deepwater's Project Specification.





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Requirements Analysis (BCI)

Check ID	Vendor Check Description	Status	Priority	Vendor Responsibility
IRC-001	Ambiguity: Flag subjective terms needing client quantification	Complete	High	All team members
IRC-002	Testability: Ensure each requirement has clear verification method	Complete	Critical	All team members
IRC-003	Consistency: Identify conflicts across PS sections	Complete	High	Ridwan (Lead)
IRC-004	Completeness: Gap analysis against maritime standards	Complete	Medium	Zong Han (Safety)
IRC-005	Quantification: Transform constraints into measurable parameters	Complete	Critical	Yuhao (Communications)
IRC-006	Interface: Verify all external interfaces defined	Complete	High	Mikhail (Autonomy)
IRC-007	Safety: Validate emergency protocols are testable	Complete	Critical	Zong Han (Safety)
IRC-008	Data: Ensure logging and retention requirements specified	Complete	Medium	Kannan (Data)
IRC-009	Communication: Confirm bandwidth/latency specifications	Complete	Critical	Yuhao (Communications)
IRC-010	Autonomy: Validate mission execution parameters	Complete	High	Mikhail (Autonomy)

Table 3: Vendor IRC Checklist Implementation

Vendor Check Categories with Client Examples:





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1. Ambiguity Detection (Our Most Frequent Finding)

- **Vendor Check:** Does requirement contain subjective terms that need client clarification?
- **Client Example:** "Secure communication" in Section 7.3
- **Vendor Action:** Flagged for client specification during communications session
- **Client Resolution:** Specified AES-256 encryption for all telemetry data

2. Testability Validation (Critical for Acceptance Testing)

- **Vendor Check:** Can we verify this requirement through testing or inspection?
- **Client Example:** "System shall be reliable" in multiple sections
- **Vendor Action:** Requested client to define reliability metrics
- **Client Resolution:** "System shall maintain 99.9% uptime with MTBF >1000 hours"

3. Consistency Checking (Preventing Implementation Conflicts)

- **Vendor Check:** Do requirements conflict across different PS sections?
- **Client Example:** Different emergency thresholds in Sections 6.4 and 7.4
- **Vendor Action:** Presented conflict to safety team for resolution
- **Client Resolution:** Unified to single specification with escalation thresholds

4. Completeness Verification (Our Domain Expertise Applied)

- **Vendor Check:** Are all necessary deep-sea operational requirements present?
- **Client Example:** Missing sensor interface protocols in Section 2.1
- **Vendor Action:** Added based on our experience with deep-sea sensor integration
- **Client Resolution:** Approved ROS2-based interface specifications we proposed





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Vendor Quantitative Results:

- Total requirements analyzed: 57
- Incongruities identified: 15 (26.3%)
- Resolved with client input: 13 (86.7% resolution rate)
- Critical findings requiring immediate attention: 2 (13.3% of incongruities)

Vendor Value Demonstrated: Our IRC process identified critical ambiguities early, preventing potential rework costs estimated at \$350K+ and reducing project risk significantly.





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4 Risk Analysis & Mitigation

Vendor Risk Assessment: As the system implementation vendor, we've identified and proactively addressed risks that could impact project success, budget, or schedule.

4.1 Critical Technical Risks (Vendor Implementation Focus)

Risk 1: Communication Specification Ambiguity

- **Vendor Identification:** PS lacked quantitative bandwidth/latency specifications we need for implementation
- **Impact on Vendor:** System failure under actual conditions could lead to costly rework (Severity: High)
- **Likelihood as Vendor:** Probable (72%) based on our deep-sea project experience
- **Vendor Mitigation:** Used Data Flow Diagram in client session to establish 1kbps, 30s requirements
- **Vendor Validation:** Tested requirements against our acoustic communication models

Risk 2: Emergency Protocol Definition Gap

- **Vendor Identification:** Missing operational procedures we need to implement
- **Impact on Vendor:** Inconsistent implementation could cause safety issues (Severity: Critical)
- **Likelihood as Vendor:** High (68%) given safety-critical nature
- **Vendor Mitigation:** Used State Transition Diagram to develop detailed protocols with client
- **Vendor Validation:** Protocols reviewed by our safety engineering team





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Risk 3: Sensor Integration Complexity

- **Vendor Identification:** Missing interface specifications we need for development
- **Impact on Vendor:** Integration delays could impact schedule (Severity: High)
- **Likelihood as Vendor:** Very Probable (85%) based on sensor specification review
- **Vendor Mitigation:** Proposed and gained client approval for ROS2-based interface standards
- **Vendor Validation:** Prototyped with highest-risk sensor types

4.2 Project Management Risks (Vendor Delivery Focus)

Risk 4: Scope Creep Post-Clarification

- **Vendor Concern:** Client clarification introduces new requirements impacting budget/schedule
- **Vendor Mitigation:** Formal change control process with requirement traceability matrix
- **Vendor Process:** All new requirements go through impact analysis before acceptance

Risk 5: Stakeholder Knowledge Attenuation

- **Vendor Concern:** Key client stakeholders become unavailable during development
- **Vendor Mitigation:** Comprehensive documentation including all clarification decisions
- **Vendor Process:** Knowledge transfer sessions scheduled with backup stakeholders





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4.3 Vendor Risk Mitigation Strategy

1. **Proactive Identification:** BCI Breakdown phase flags risks before implementation begins
2. **Early Client Engagement:** Clarify phase addresses risks through direct stakeholder discussion
3. **Continuous Monitoring:** Risk register maintained and reviewed weekly
4. **Escalation Protocols:** Defined pathways for risks exceeding agreed thresholds

Vendor Risk Management Value: Our proactive risk identification and mitigation approach reduces project uncertainty and provides clear visibility into potential challenges before they impact delivery.





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5 Account Manager & Stakeholder Coordination

5.1 Vendor Account Manager Role

- **Name:** Muhammad Ridwan Putra Jasni
- **Vendor Role:** Primary Brigman Deep Sea Technology representative to Triton Deep-water
- **Responsibilities:**
 - Facilitate all client-vendor communications and coordination
 - Schedule and lead all clarification sessions with appropriate stakeholders
 - Manage client expectations and ensure alignment on requirements
 - Escalate issues through proper channels when needed
- **Vendor Value:** Ensures single point of contact for consistent messaging and relationship management

5.2 Vendor Stakeholder Engagement Strategy

- **Scheduled Sessions:** 3 targeted clarification sessions with specific stakeholder groups
- **Visual Aids Deployed:** Used models as discussion tools to bridge technical-operational gaps
- **Vendor Documentation:** Comprehensive records of all client decisions and clarifications
- **Follow-up Process:** Action items tracked to completion with client validation





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5.3 Vendor Escalation Protocol

1. **Level 1:** Vendor technical team attempts resolution (within 24 hours)
2. **Level 2:** Account Manager mediates between teams (within 4 hours if unresolved)
3. **Level 3:** Vendor executive engagement (VP of Engineering contacts client counterpart)
4. **Level 4:** Contractual and scope review with legal teams

5.4 Vendor Coordination Outcomes

- **86.7% Resolution Rate:** Successfully resolved majority of incongruities with client input
- **Strengthened Client Relationship:** Professional engagement built trust for ongoing collaboration
- **Clear Communication Channels:** Established protocols for efficient future communications
- **Mutual Understanding Achieved:** Bridged client operational needs with our technical capabilities

Vendor Relationship Management Success: Our structured approach to stakeholder coordination has established a strong foundation for successful project delivery and potential future engagements.





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6 Declarations

Vendor AI Tool Usage Declaration:

- ChatGPT was used as a tool to assist with language refinement, grammar checking, and improving document flow.
- The AI tool served as an educational resource to help explain complex technical concepts within our vendor team.
- All analysis, models, diagrams, and substantive content in this report were developed by the Brigman Deep Sea Technology analysis team.
- AI-generated content was critically evaluated, validated against our technical expertise, and substantially modified to ensure accuracy and relevance to the project.

Vendor Originality Declaration:

- This report represents original analysis work conducted by the Brigman Deep Sea Technology team as the selected vendor.
- The BCI methodology is a proprietary framework developed and applied by our organization.
- All visual models and analysis techniques were created specifically for the Abyss Navigator Suite project requirements.
- Quantitative findings and metrics are based on our actual analysis of Triton Deepwater's Project Specification.

Vendor Compliance Declaration:

- This analysis complies with IEEE 830 standards for requirements specification as required for SRS development.
- All requirements meet CRaM principles (Correct, Relevant, Measurable) ensuring





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testable specifications.

- The analysis follows industry best practices for requirements engineering from a vendor perspective.
- Traceability is maintained from Project Specification through client clarification to interpreted requirements.

Sign-off & Repository Information

VENDOR REPOSITORY INFORMATION

<https://github.com/Spooky312/The-Abyss-Navigation-Suite-INF2113>

Vendor Repository Organization:

- **/requirements/** – PS, BCI analysis, interpreted specifications
- **/models/** – Context, State Transition, Data Flow, Use Case diagrams
- **/client-sessions/** – Meeting minutes, decisions, action items
- **/artifacts/** – IRC checklists, risk registers, validation documents

All artifacts version-controlled for client transparency and auditability

VENDOR TEAM APPROVAL SIGN-OFF

Vendor Team Member	Employee ID	Signature
Muhammad Ridwan Putra Jasni	BDST-RE-001	_____
Chng Zong Han	BDST-SE-015	_____
Lin Yuhao	BDST-CS-022	_____
Muhammad Mikhail Bin Mazlan	BDST-AI-029	_____
Kannan s/o Rajamohan	BDST-DM-036	_____





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Requirements Analysis (BCI)

Vendor Report Version: 2.0 (BCI Methodology)

Client Delivery Date: February 10, 2026

Vendor Status: Analysis Complete - Ready for Client Review and SRS Development

Next Client Step: Review and approval of interpreted requirements and specifications

